(ii) Intensity (or) Amplitude division: If we allow light to pass through a partially silvered mirror (beam splitter), both reflection and refraction take place simultaneously. As the two light beams are obtained from the same light source, the two divided light beams will be coherent beams. They will be either in-phase or at constant phase difference as shown in Figure 7.9. Instruments like Michelson's interferometer, Fabray-Perrot etalon work on this principle.

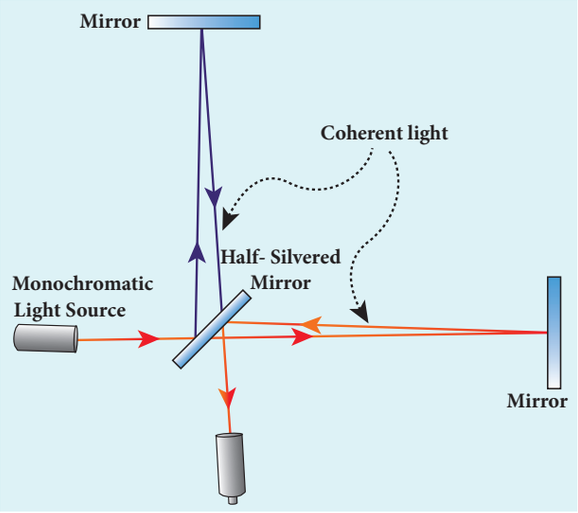


Figure 7.9 Intensity (or) amplitude division

(iii) Source and Images: In this method a source and its images will act as a set of coherent sources, because the source and its image will have waves in-phase (or) constant phase difference as shown in Figure 7.10. The Instrument, Fresnel's biprism uses two virtual images of the source as two coherent sources and the instrument, Lloyd's mirror uses a source and its one virtual image as two coherent sources.

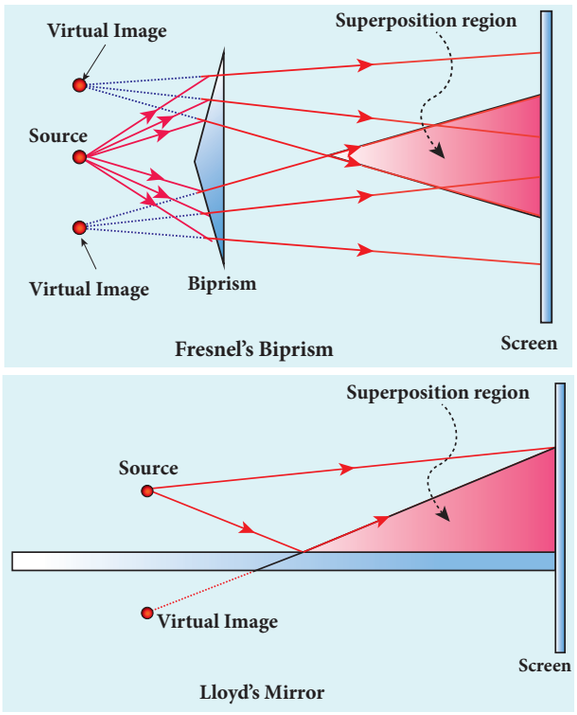


Figure 7.10 Source and images as coherent sources

7.3.3 Double slit as coherent sources

Double slit uses the principle of wavefront division. Two slits S_1 and S_2 illuminated by a single monochromatic source S act as two coherent sources. The waves from them travel in the same medium and superpose. The constructive and destructive interference formed by them are shown in Figure 7.11(a). The crests of the waves are shown by thick continuous lines and troughs are shown by broken lines in Figure 7.11(b).

At points where the crest of one wave meets the crest of the other wave (or) the trough of one wave meets the trough of the other wave, the waves are in-phase. Hence, the displacement is maximum and these points appear bright as a result of this ***constructive interference***.

At points where the crest of one wave meets the trough of the other wave and

XII_Physics_Vol-2_UNIT-7_EM.indd 62 03-03-2022 15:18:45